

Foundation models for atomistic simulation of chemistry and materials

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Abstract

Conventional computational methods for modeling chemical and materials systems are limited by system size and timescale, forcing a trade-off between quantum-mechanical accuracy and the sampling needed for realistic observables. Large language and vision foundation models – pre-trained on massive datasets using transformer architectures – have revolutionized many fields. It is thus interesting to ask whether a foundation model – subject to suitable data, parameter scaling and training – could enable learned simulations of chemistry and materials. Here, we review the field of machine-learned interatomic potentials (MLIPs) and posit that scaling up large and diverse chemical and materials datasets and highly expressive architectures using advanced training strategies should result in models that are: more efficient, transferable, robust to out-of-distribution scenarios, and easier to fine-tune to a variety of downstream physical observables than models trained from scratch on small datasets corresponding to specific, targeted atomistic simulation tasks. We provide specific criteria for creating such large-scale MLIP foundation models, coordinated strategies for their development, evaluation and deployment, and highlight potential emergent capabilities that could transform predictive simulations in chemistry and materials science and accelerate discovery across multiple technological domains.

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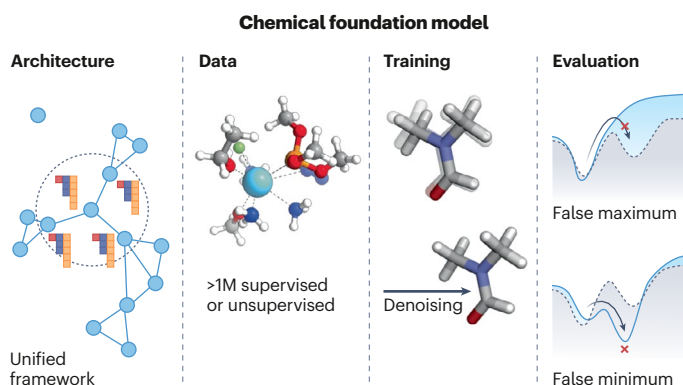
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Introduction

The theoretical foundation for describing atomistic simulations in chemistry and materials science is the Schrödinger equation. This allows a chemical system and its physical properties to be uniquely defined by the Cartesian and spin coordinates of the constituent atoms, in addition to the net charge of the system. However, as famously stated by Dirac after the discovery of quantum mechanics, “the underlying physical laws necessary for the whole of chemistry are thus completely known, and the difficulty lies only in the fact that the exact application of these laws leads to equations much too complicated to be soluble”. This observation led to a series of well-controlled approximations – such as the Born–Oppenheimer separation of nuclei and electronic degrees of freedom and the transformation of the Schrödinger eigenvalue equation into an algebraic framework – allowing good model chemistries to be solved for systems beyond the hydrogen molecule¹. It is equally important to satisfy the laws of statistical mechanics, which require high-dimensional integration over Boltzmann-weighted configurations to predict measurable physical properties. This is usually achieved by sampling over time-dependent trajectories, with the hope that ergodicity is satisfied (that is, that all relevant configurations are captured by observing the system long enough)². Here, a fundamental tension emerges: quantum-mechanical accuracy and statistically converged sampling are mutually exclusive under current computational limits. Even the most widely used approximate electronic structure methods based on density functional theory (DFT)³ combined with ab initio molecular dynamics remain computationally prohibitive for the large timescales and large length scales and system sizes that define many of the most interesting areas of chemistry and materials science.

To overcome this tension, we can leverage the tools of artificial intelligence and deep learning. Their success in other domains suggests that we could learn ‘the foundations’ of molecular and materials chemistry from an abundance of data using appropriate model architectures and training strategies. The concept of a foundation model has been established by the natural-language-processing and computer-vision communities through the development of large language models (LLMs) and large vision models (LVMs)^{4,5}. These models are pre-trained in an unsupervised or self-supervised manner on enormous amounts of data, and they can then be easily fine-tuned for accurate prediction on a variety of downstream tasks, revolutionizing the field of artificial intelligence.

Although LLMs and LVMs themselves hold great potential to affect chemistry and materials science^{6–10}, in this Perspective, we explore whether foundation models can be created to directly capture the potential energy surface (PES) via machine-learned interatomic potentials (MLIPs)^{11,12}. A foundation model targeting the PES would enable large-scale atomistic simulation and analysis, including the ability to derive any observable of interest, for any molecular or periodic system, with minimal¹³ or no system-specific fine-tuning. Importantly, it would supersede the painful acquisition of new labelled data for each system of interest, and the hidden expense of active learning, which can require hundreds of rounds of retraining to develop a robust MLIP^{14–17}. The end goal of such a model is to enable large-scale atomistic simulation and analysis, including the ability to derive any observable of interest, for any molecular or periodic system, either with a small amount of system-specific fine-tuning¹³ or without any fine-tuning at all.

To help guide the community towards such a goal, this Perspective provides clear definitions of a foundation model for atomistic simulation of chemistry and materials, and it explains how one applied for atomistic simulation should be distinguished from a ‘universal potential’

or more narrowly defined concepts such as transfer learning. Our criteria begin with the ability of a MLIP foundation model to leverage pre-training on large datasets and exhibit consistent scaling laws over many orders of magnitude of data and training compute, strategies that have made LLMs and LVMs so powerful. Within this context, we assess current model architectures and identify limitations and opportunities for their scalability to larger datasets and amount of learnable parameters.

We also consider the type and amount of data required for a generally applicable pre-trained foundation model, how that compares with existing data, and where we expect fine-tuning data to be most important for chemistry and materials science applications. Although our focus is on foundation models trained on energy and force data (which directly define the PES) for molecular simulation, we acknowledge the existence of different modalities of data that could also inform a foundation model to address outstanding problems in chemistry and materials science. These include images and text for catalytic discovery¹⁸ and the use of video data such as that generated for liquid-phase transmission electron microscopy imaging¹⁹.

In addition, we examine the importance of advanced training strategies, including self-supervised and unsupervised pre-training, as well as model distillation, and highlighting that new methods for physical infusion must be both relevant and feasible for large-scale training on advanced hardware. Moreover, we consider how the performance of MLIPs is currently evaluated, and we suggest new benchmarks for demonstrating scaling laws, molecular dynamics stability, and the ability to be broadly fine-tuned for a wide range of downstream applications, which is particularly important given the breadth of observables in chemical and materials systems. We conclude with a summary and future directions, including the potential impact in terms of real-world applications, and the societal and ethical issues that must also be considered in the development and deployment of MLIP foundation models.

Characteristics of foundation models

The term ‘foundation model’ has been defined to mean any large-scale model pre-trained on large amounts of diverse data to capture a broad range of complex patterns⁵, as represented by popular LLMs, such as GPT²⁰ and Llama²¹. One of their defining characteristics is their tendency to obey empirical scaling laws^{4,22–24}, with performance improving as a function of increasing model size, training data size, and computational resources. Increasing model capacity improves expressivity and performance, especially when coupled with improved algorithms for neural network operations, such as the attention mechanisms most commonly used in highly scalable Transformer²⁵ architectures. Data scaling for pre-training refers to increasingly large and diverse datasets of typically unlabelled data to learn general features and patterns, covering various domains and contexts. Compute scaling permits parameter and data scaling through optimal use of graphics processing unit (GPU) and central processing unit hardware, exploiting single and mixed precision computation when appropriate, and taking advantage of parallelization using distributed computing. These scaling laws have been crucial for the success of foundation models across various domains, including natural language processing, computer vision, climate modelling, and robotics^{26–29}. In fact, LLMs have exhibited emergent properties in chemistry for generative tasks in the design⁸ and synthesis¹⁰ of small-molecule drugs when fine-tuned on the syntax of SMILES strings (that is, text-based encodings of molecular structures) and chemical reaction templates.

In addition to scaling laws, foundation models rely on effective pre-training strategies that enable strong performance on benchmark datasets and later fine-tuning for downstream tasks. Pre-training objectives – particularly in language models – often involve self-supervised learning tasks, such as predicting missing words in sentences or predicting the next word in a sequence. Earlier in the training process, more compute is used, data quality is typically lower, and human supervision is minimal. Scaling the model, data and compute can lead to emergent capabilities, wherein a foundation model becomes capable of solving tasks that were not possible at smaller scales³⁰. It is still disputed how and why these capabilities are unlocked at scale³¹. However, empirical scaling laws provide a quantitative framework for understanding and predicting the behaviour of chemical foundation models as they are scaled up, and they offer insights into how different factors (such as architecture, data and pre-training) contribute to improved model performance.

An important clarification that we hope to address in this Perspective is to distinguish the term foundation model from related terms, such as ‘transferable’ or ‘universal’ MLIPs. We offer that a true foundation model should demonstrate some of the following features. First, a fine-tuned MLIP foundation model should perform across a broad range of downstream tasks, outperforming task-specific models trained from scratch^{32,33}. Second, it should comply with heuristic scaling laws – showing improved performance with increasing training data size and more parameters, and increasing compute resources. Last, it should demonstrate emergent capabilities, such as generalizing beyond the training domain or predicting properties very different from those contained in the underlying training data (for example, predicting higher-quality coupled-cluster singles and doubles with perturbative triples (CCSD(T)) data or magnetic field-dependent properties).

At present, no MLIPs are true foundation models according to these criteria. For example, the popular MLIP MACE-MP-0 (ref. 34) is trained on data from the Materials Project^{35,36} generated at the Perdew–Burke–Ernzerhof (PBE) level of DFT. The resulting machine learning model exhibits good accuracy in energy and force predictions against PBE on other crystalline unit cells and exhibits excellent stability in the numerical solution to Hamilton’s equation of motion across related chemical and materials systems. This has opened up a whole new range of applications, including free energy calculations, simulating spectroscopic observables, and explorations of phase diagrams. However, we would stipulate that MACE-MP-0 is not a foundation model but an excellent universal PBE potential, given that it is trained with supervision to do a very specific task: predict energy and force labels for a single chemical domain at one level of theory. Another key limitation is that MACE-MP-0 (ref. 34), like many universal MLIPs, does not attempt to distinguish between systems with different total charge and spin, whose energy and forces would be different although the underlying input structure is the same.

In the remainder of this Perspective, we consider the necessary ingredients for a foundation model MLIP for atomistic simulations, such as those now mainstream for language and vision modalities. These ingredients are illustrated in Fig. 1. We posit that training such a model will require the following: neural network architectures that either incorporate physical inductive biases or learn these biases directly from the data – and in either case, maintain consistent performance improvements as model size and training compute are increased; generating ample training data with wide coverage across the periodic table and over a large range of physical conditions; and

pre-training and fine-tuning strategies that help the MLIP foundation model to unlock improvements at scale. With sufficient computational resources and rigorous validation by the community, we are optimistic about the ability of foundation MLIP models to transform atomistic simulation in chemistry and materials science.

Machine learning architectures for chemistry and materials

The architectures of MLIPs are distinct from those developed in domains such as natural language processing, primarily to incorporate various physical inductive biases relevant to chemical and material systems. The exact manner of realizing the physical constraints has gone through several waves of evolution. We provide a general survey of these developments, as well as an outlook on recent architectures that increasingly exploit data and parameter scaling as a driving factor in performance.

Evolution of MLIP architectures and inductive biases

Over the past two decades, MLIP architectures have evolved substantially, integrating the inductive biases of chemistry (that is, assumptions or constraints that reflect known chemical principles) while incorporating advancements from other areas of machine learning. The early Behler–Parrinello neural network potentials³⁷ used hand-crafted two-body and three-body descriptors for describing atomic environments. Later, DeePMD models³⁸ automated the optimization and discovery of such descriptors, reducing reliance on manual feature engineering. The atomic cluster expansion (ACE) method³⁹ introduced a unified and generalizable framework for constructing atom-centred descriptors using systematically improvable body-ordered terms. Graph neural networks then broadened the landscape, allowing for flexible and expressive architectures that learn features by iteratively exchanging information between neighbouring atoms, thus capturing many-body effects^{40–43}. More recently, graph neural networks (GNN) models have incorporated different forms of graph attention mechanisms^{44,45}. These models can be viewed as a more general way of performing message passing between atoms.

Thus, the diverse existing architectures are seen to share the same mathematical foundation. In GNNs, chemical and material systems are represented as graphs, wherein atoms are represented as nodes, and connecting edges are constructed based on a distance cut-off. Additionally, almost all atomic and structural representations for materials and molecules are fundamentally related to and can be understood within the unifying framework of atomic density^{46,47} or, equivalently, ACE³⁹. Moreover, these representations can also be viewed as specific graph features, encoding geometric and chemical information of the nodes. This duality highlights a deeper connection: most existing MLIPs can be understood as different variations of the same underlying framework.

Most MLIPs incorporate three inductive biases that are specific to chemical and materials systems. The first is locality or ‘nearsightedness’. This postulates that energy and forces experienced by a central atom are primarily influenced by neighbouring atoms within a finite cut-off radius. Locality ensures the key model chemistry requirement of size consistency¹ is satisfied. Message-passing schemes relax this assumption by enabling atoms to aggregate information from progressively more distant neighbours across multiple iterations, although nearby neighbours still contribute the most. The second bias relates to physical constraints, such as energy conservation, smoothness and invariances (translational, rotational and permutational).

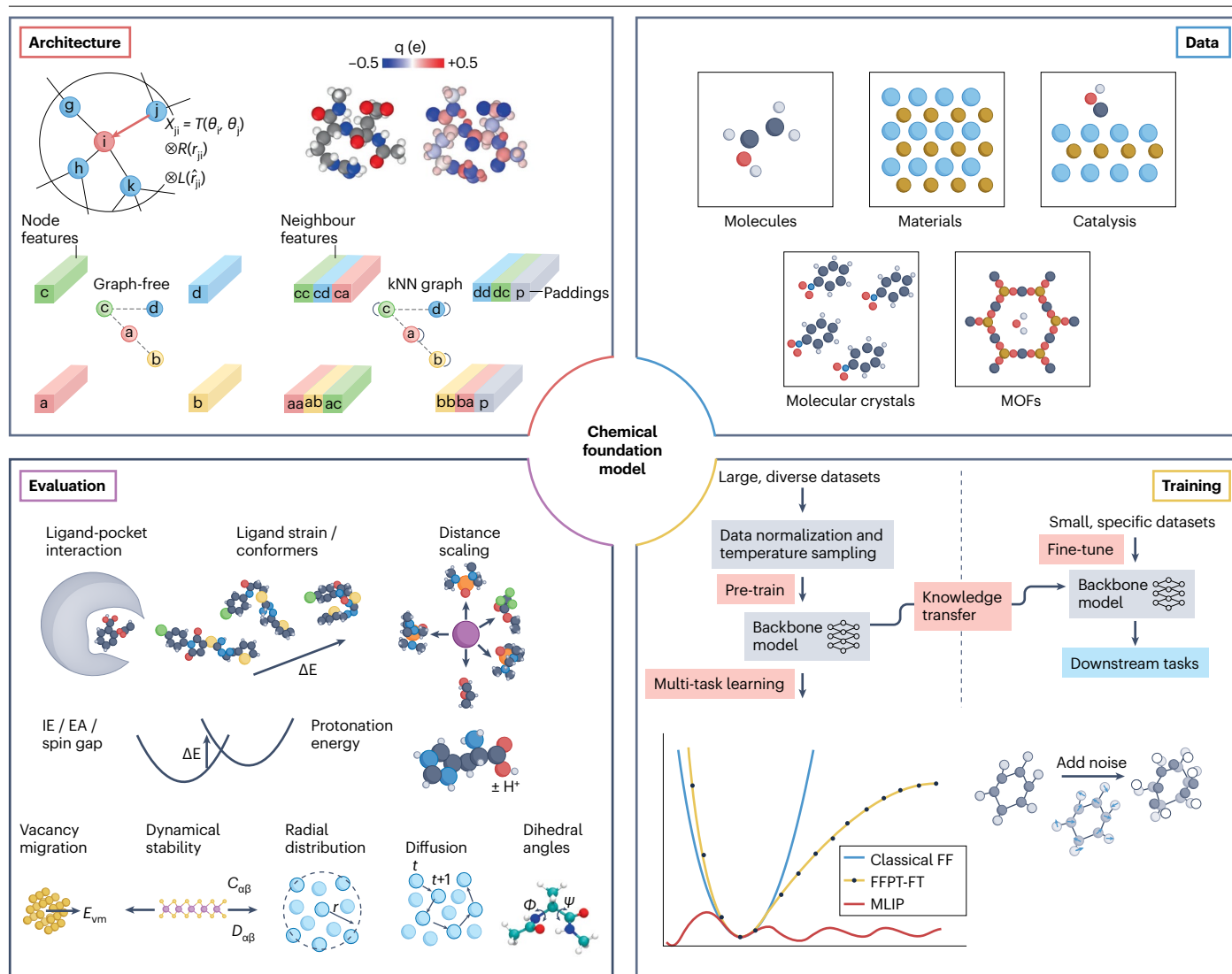


Fig. 1 | Overview of foundation models for machine learning interatomic potentials. Upper left: Various aspects of the MLIP architecture. Upper right: Various aspects of the MLIP training data. Lower right: Various aspects of the MLIP training method. Lower left: Various aspects of the MLIP evaluation. EA, electron affinity; IE, ionization energy; kNN, k -nearest neighbours; MAE, mean absolute error; MLIP, machine-learned interatomic potential; MOF,

metal–organic framework. Elements of the upper left panel are adapted and reprinted from refs. 198,199, respectively, Springer Nature Limited, and adapted from E. Qu et al. (manuscript in preparation) under a Creative Commons licence CC BY 4.0. Elements of the lower right panel are adapted from ref. 139 under a Creative Commons licence CC BY 4.0. Elements of the lower left panel are adapted from refs. 164,175 under a Creative Commons licence CC BY 4.0.

These constraints are incorporated either explicitly within the model architecture (for example, through gradient-based forces, symmetrized input features, or invariance-preserving transformations) or implicitly via the loss function and training data. The third bias relates to the description of interatomic interactions via body-order terms. Two-body (pairwise) terms dominate, but three-body terms are also important. Higher-body terms provide diminishing returns owing to increased computational cost and reduced relative contributions. Frameworks such as ACE³⁹ provide systematic control over the inclusion of higher-body terms, with user-specified truncation. In GNNs, adding a message-passing layer effectively increases the body order by one to yield models of different ranks.

Scaling and the ‘bitter lesson’

The unifying framework of current MLIP architectures involves enforcing physical constraints. This stands in contrast with the well-known bitter lesson, which argues that general methods that leverage computation and data are more effective in the long term over approaches that try to build knowledge into the machine learning pipeline⁴⁸. Constrained models face challenges such as unstable training dynamics arising from ill-conditioned loss landscapes, as well as difficulty with parallelization and distributed training on modern GPU hardware^{49–51}. Indeed, we are now seeing that methods eschewing physical constraints in favour of scalable and expressive architectures can outperform models with built-in constraints and learn these constraints

directly from data^{28,29,45,52–54}. These models are often significantly more compute-efficient owing to the use of highly optimized dense primitive operations such as attention²⁵. This offers the first glimpse of the bitter lesson in the context of an MLIP architecture: models with fewer inductive biases may eventually perform better than those with more biases, as observed in other areas of machine learning. Moreover, this suggests that models that leverage data and parameter scaling – rather than those with built-in constraints – will be more effective for chemical and materials systems.

Although simple physical constraints are easily enforced in machine learning models, whether more complex physical laws can be learned from data or still require inductive biases remains an open question. For example, rotational equivariance^{42,43,55} can be approximately enforced via data augmentation (that is, rotational transformations of the existing data) in nearly all cases^{45,46,56}. Whether more complicated physical laws can be learned from data alone or still require some inductive biases remains an open question. An example would be preserving the symplectic structure of Hamilton's equation of motion – required for stable molecular dynamics simulations. In this case, stability in molecular dynamics simulations requires two key conditions. The first condition is the absence of pathological behaviours, such as molecules 'exploding' owing to data insufficiencies and numerical instabilities,

even at very small time discretizations (Fig. 2a). The second condition is the ability to perform molecular dynamics simulations without introducing non-conservative and energy drift artefacts. Energy conservation is agnostic to the MLIP architecture itself, as it is typically enforced in gradient-based force models by taking gradients of the predicted potential energy in the loss function – an expense that is avoided in direct force models^{55,56}. Recent studies have observed that certain MLIPs that directly predict non-conservative forces can induce artificial heat generation in NVE simulations or severely modified dynamics in NVT simulations⁵⁷. These behaviours undermine the reliability of such non-conservative MLIPs for long-timescale dynamics and their integration into predictive downstream workflows. Even so, these undesirable behaviours can also be addressed with data-driven post-training strategies discussed later.

Current limitations and challenges

Current MLIP architectures have several limitations that should be examined more closely. The first limitation is transferability outside the training data. Although recently developed universal MLIPs demonstrate reasonable performance in predicting materials stability on element-wise out-of-distribution datasets^{58,59}, these evaluations primarily focus on local energy minimum searches through structure

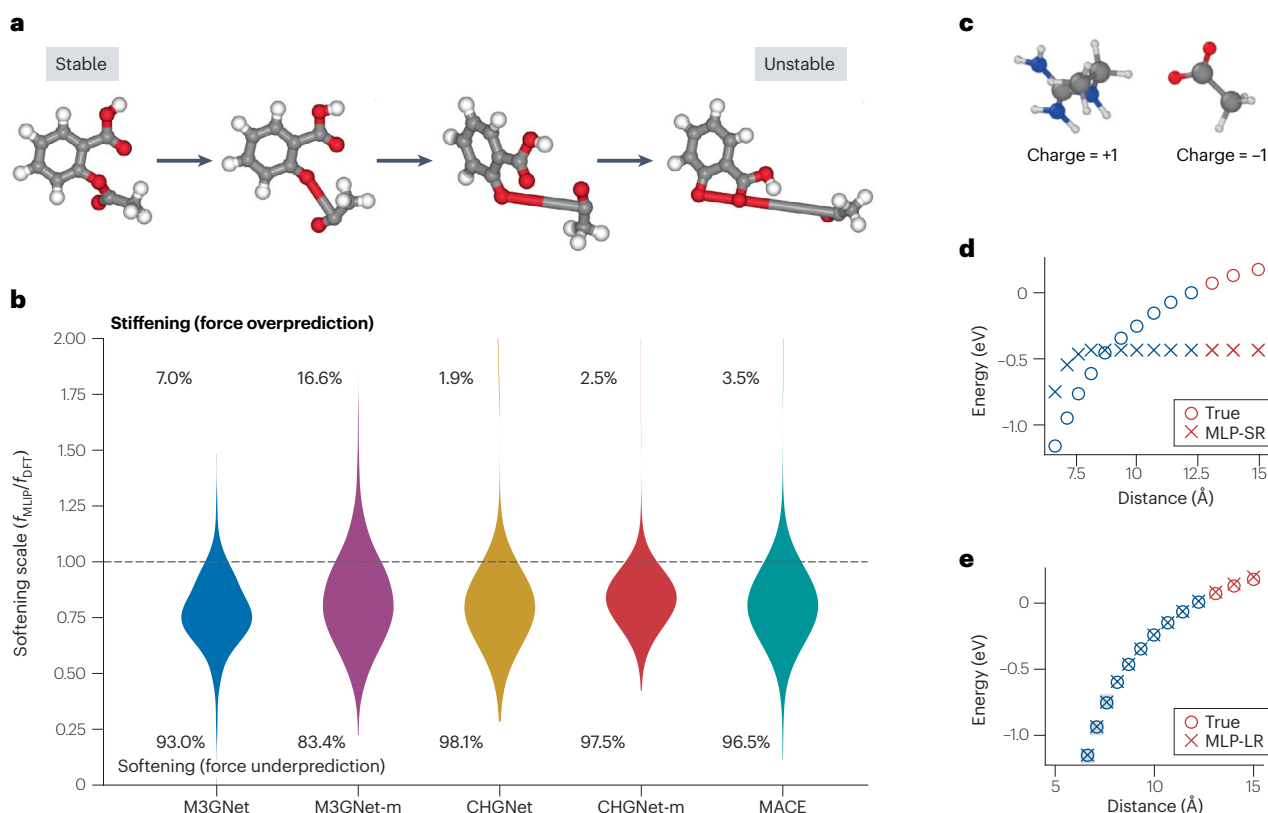


Fig. 2 | Current advances in MLIPs for molecular simulation. **a**, Depiction of instability in molecular simulations with MLIPs, which can occur even at very small time discretizations¹³³. **b**, Distribution of softening scales ($f_{\text{MLIP}}/f_{\text{DFT}}$) sampled from 1,000 high-energy configurations using the structures from the crystalline compound dataset⁵⁸. The horizontal dashed line means no softening and no hardening. **c**, An example of two charged molecules that are far apart and are predicted to be non-interacting. This non-interaction is because of short-range implicit bias from graph neural networks and message passing

(part **d**) and can be overcome using a machine learning model that incorporates long-ranged information (part **e**). The blue points represent training and the red points represent testing. DFT, density functional theory; MLIP, machine learning interatomic potential; MLP-LR, long-range machine learning potential; MLP-SR, short-range machine learning potential. Part **a** is reprinted from ref. 133 under a Creative Commons licence CC BY 4.0, part **b** is reprinted from ref. 61, and parts **c–e** are adapted from ref. 70, Springer Nature Limited.

relaxation of crystalline materials, sharing similar prototypes in the Materials Project database. A systematic softening (underestimation) of the machine learning-predicted PESs and interatomic forces⁶⁰ compared with DFT ground truth was observed in a recent benchmark study⁶¹ (Fig. 2b). The underestimation of the PES is particularly significant for high-energy atomic configurations, which limits its application to molecular dynamics simulations and rare-event sampling. Recent large data efforts^{32,62} and open-sourced universal MLIPs^{63,64} that include more off-equilibrium atomic configurations show promise in addressing these challenges, as discussed in the following section.

Another limitation is the lack of incorporation of long-range interactions. Although short-range potentials may be sufficient to describe most properties of homogeneous bulk systems⁶⁵, they may fail for interfaces⁶⁶, dielectric response^{67,68} and dilute ionic solutions with Debye–Hückel screening. For example, consider two charged molecules that are separated by a distance greater than the cut-off radius used in most MLIPs (Fig. 2c). These models, which focus on short-range interactions, will fail to accurately describe the electrostatic forces between the molecules (Fig. 2d). This issue cannot be resolved simply using message passing because the molecules exist on separate graphs and do not interact within the framework of short-range biases inherent to message-passing neural networks. By contrast, MLIPs with built-in long-range corrections^{69–71} can effectively solve such artefacts (Fig. 2e), although whether such strategies will still be effective when training at scale on very large datasets remains an open question. Models with global attention mechanisms may also be able to learn long-range interactions in a data-driven manner⁷².

There is also the concern of scalability, specifically whether scaling laws will be exhibited as the community acquires larger and larger datasets and computational resources. Is the GNN architecture adequate or is there a need for more scalable architectures^{72–74}? Relatedly, computational inefficiency when applying MLIPs at scale⁷⁵ can limit their application to realistic chemical and materials problems. The current short-ranged MLIPs are already orders of magnitude more expensive than traditional force fields. Further incorporating long-range interactions may incur more computational cost. There is a willingness to pay this price for improved accuracy; however, the current MLIP frameworks are not fully optimized for such tasks. In recent work, Kreiman and co-workers used unmodified Transformers trained directly on Cartesian coordinates – without predefined graphs or physical priors – and demonstrated comparable mean absolute errors for energy and forces relative to a state-of-the-art equivariant GNN, while being faster in both training and inference speeds⁷². Such architectures were shown to be scalable and, thus, respect empirical scaling laws without saturating⁷², and offer new possibilities for foundational MLIPs.

Data requirements, pitfalls and challenges

Real-world image and language datasets often contain trillions of tokens, whereas simulated chemistry datasets for training MLIPs typically contain between 1 to 200 million molecule or material configurations with associated energy and force data. Thus, increases in dataset size remain crucial for improving the pre-training of MLIPs if they are to ultimately serve as robust foundation models. In this section, we consider the existing public datasets for training foundation MLIPs, assessing their limitations and areas wherein data coverage is still incomplete.

Size, composition and diversity of datasets

Table 1 summarizes molecular and materials datasets with over 1,000,000 energy-labelled and force-labelled structures that have

emerged over the past ~5 years. The underlying labelling methods are DFT calculations with variable density functionals and basis sets. We also include large-scale non-DFT chemical data, such as Uni-Mol2 (ref. 76) (a curated subset of the Zinc20 (ref. 77) molecular dataset), and Zinc22 (ref. 78) (comprising 4.5 billion molecules with force field-derived labels of their physical properties, including partial atomic charges, cLogP values and solvation energies). In all cases, this variable quality in data is appropriate for pre-training, wherein more value should be placed on broadly covering chemical space. This includes domain diversity (for example, molecular, interfacial and solid-state systems); elemental diversity; diversity in both total charge and spin; molecular and structural diversity, covering possible bonding and interaction motifs; and configurational diversity, including both near-minima and far-from-minima configurations.

For covering the periodic table, most plane-wave DFT datasets follow the standards set by the Materials Project^{35,79} and, thus, represent the chemical elements up to plutonium. By contrast, molecular force field and DFT datasets are considerably more constrained in their elemental coverage, with most datasets only including C, H, N and O and some coverage of S, F, Cl and Br. The only molecular datasets of over 1 million data points with both energy and forces labels and extend beyond this set of elements are SPICE⁸⁰ (P, Li, Na, Mg, K, Ca and I) and AIMNet2 (ref. 81) (P, I, Si, B, As and Se), and more recently QCML⁸² (73 elements total) and Open Molecules 2025 (OMol25)⁸³ (83 elements).

Related to elemental diversity are the atomic properties of charge and spin that are important chemical features of atoms in molecules. The only large-scale datasets with variable-charge species are SPICE(2)⁸⁴, AIMNet2 (ref. 81), AIMNet-NSE⁸⁵, solvated protein fragments⁸⁶, GEMS⁸⁷, QCML⁸² and OMol25 (ref. 83). Although the AIMNet-NSE⁸⁵ dataset contains variable spin species, it carries no force labels, whereas QCML does include both labels for some limited spin species⁸², although variable spin is much better covered by OMol25 (ref. 83). Chemical diversity also depends on the variety of chemical interactions captured by the data. Although most of the historical molecular datasets mentioned thus far focus on isolated molecules, SPICE⁸⁰ captures varied intermolecular interactions between molecules, including some metal ions; solvated protein fragments⁸⁶ and GEMS⁸⁷ include protein fragments with multiple nearby solvent species; and ANI-2x⁸⁸ includes dimer scans and water clusters. Recognizing the value of intermolecular interactions, OMol25 also includes these data types as well as comprehensive sampling of molecular clusters across biomolecules, electrolytes, and metal complexes⁸³.

To create structural diversity, large-scale data-acquisition efforts use a collection of sampling techniques – including normal mode analysis, ab initio and empirical molecular dynamics, minimization and relaxation, reaction path sampling, metadynamics and active learning – as outlined in Table 1. Structural diversity also encompasses different regions of the PES covered by the dataset. Ensuring a broad range of forces between 0 and -10 eV Å^{-1} is important to effectively learn chemically important components of the PES encountered during molecular dynamics simulations, geometry optimization, and reaction path and transition state optimization. Until recently, Transition-1x⁸⁹ was the only large-scale dataset that explicitly samples reaction paths and metastable structures such as transition states – crucial areas for any truly foundational MLIP^{90,91}. Extremely high forces ($>50\text{ eV Å}^{-1}$) can be detrimental to training and are usually filtered out, although high-energy and force data can be weighted with a Boltzmann factor¹⁷, which still provides the model important chemical information about these regions.

Table 1 | Publicly available large-scale datasets

Datasets	Size	Calculation method(s)	Sampling method(s)
Molecular datasets			
OMol25 (ref. 83)	100M	ω B97M-V/def2-TZVPD	Molecular dynamics
			Rattling
			Geometry optimization
			Reaction pathways
			ANI-2x ⁸⁸ , OrbNet Denali ⁹²
			Transition-1x ⁸⁹ , SPF ⁸⁶
			SPICE ⁸⁰ /SPICE2 (ref. 84)
QCML ⁸²	33.5M	PBE0-D4/NAOs ¹⁹²	Chemical graphs
			Normal mode
OMC25 (ref. 182)	27M	PBE-D3/PAW	Relaxation trajectory
AIMNet2 (ref. 81)	20M	ω B97M-D3/def2-TZVPP	ANI-2x ⁸⁸ , OrbNet Denali ⁹²
			Normal mode
			Metadynamics
			Molecular dynamics
			Torsion scan
∇^2 DFT ¹⁹³	16M	ω B97X-D/def2-SVP	Relaxation trajectory
Transition-1x ⁸⁹	10M	ω B97X/6-31G*	Nudged elastic band
ANI-1x ¹¹⁸ /ANI-2x ⁸⁸	8.9M	ω B97X/6-31G*	Dimer and torsion scans
			Normal mode
			Molecular dynamics
			Active learning
QM7-X ¹⁹⁴	4.2M	PBE0-MBD	Normal mode
SPF ⁸⁶	2.7M	RPBE-D3(BJ)/def2-TZVP	Molecular dynamics
			Active learning
GEMS ⁸⁷	2.7M	PBE0-MBD/def2-TZVPP	SPF ⁸⁶
			Molecular dynamics
SPICE ⁸⁰ /SPICE2 (ref. 84)	2M	ω B97M-D3(BJ)/def2-TZVPPD	Molecular dynamics
			Dimer scan
Materials datasets			
OC20 (ref. 168)	265M	RPBE/PAW	Relaxation trajectory
			Molecular dynamics
			Rattling
OMat24 (ref. 62)	110M	110M PBE(+U)/PAW	Rattled Boltzmann
			Molecular dynamics
			Relaxation trajectory
ODAC25 (ref. 178)	60M	PBE-D3(BJ)/PAW	Relaxation trajectory
			Grand canonical Monte Carlo

Datasets	Size	Calculation method(s)	Sampling method(s)
ODAC23 (ref. 179)	38M	PBE-D3(BJ)/PAW	Relaxation trajectory
Alexandria ¹⁹⁵	30M	PBE(+U)/PAW, SCAN/PAW	Relaxation trajectory
		PBEsol/PAW	
OC22 (ref. 196)	9.8M	PBE(+U)/PAW	Relaxation trajectory
OC25 (ref. 180)	7.8M	RPBE-D3/PAW	Relaxation trajectory
			Molecular dynamics
MPtrj ³⁶	1.6M	PBE(+U)/PAW	Relaxation trajectory
Unsupervised datasets			
Zinc20 (ref. 77)/Zinc22 (ref. 78)	4.5B	MMFF94	Torsion sampling
Uni-Mol ¹⁹⁷ /Uni-Mol2 (ref. 76)	838M	MMFF94	Distance geometry
			Geometry optimization

We listed datasets of more than 1 million 3D structures with energy and force labels, as well as datasets of more than 100 million 3D structures without these labels. B, billion; M, million; OC20, Open Catalyst 2020; ODAC25, Open DAC 2025; OMat24, Open Materials 2024; OMC25, Open Molecular Crystals 2025; OMol25, Open Molecules 2025; SPICE, small-molecule/protein interaction chemical energies; SPF, solvated protein fragments.

At this point, it is appropriate to highlight the recent release of OMol25 (ref. 83), the largest molecular DFT dataset for training MLIPs which covers 83 elements of the periodic table with over 100 million snapshots. In addition to generating many new structures, OMol25 encompassed configurations from the ANI-2x⁸⁸, SPICE2 (ref. 84), Transition-1x⁸⁹, OrbNet Denali⁹² and SPF⁸⁶ datasets that were recalculated at OMol25’s excellent level of DFT (ω B97M-V/def2-TZVPD). OMol25 includes energies and forces for systems up to 350 atoms, up to ten unpaired electrons, total charges ranging from $-10e$ to $+10e$, large coverage of spin species, small molecules, electrolytes, biomolecules, metal complexes and intermolecular interactions, using more than ten distinct methods for structural sampling.

Error sources in DFT reference data

The main feature of pre-trained MLIP foundation models is that they can be fine-tuned with a modest amount of data to yield accurate PES for diverse chemical systems, or extended to predict properties beyond energies and forces. At present, DFT calculations provide the best trade-off between accuracy and computational cost, such that nearly all supervised large-scale datasets for MLIPs are composed of DFT calculations. There are at least three categories of errors in the reference values of such datasets, any one of which alone could justify the need for better fine-tuning data.

The first significant source of error is the choice of functional (or wavefunctional, if one goes beyond DFT). Standard generalized gradient approximations (GGAs) have root-mean-square (RMS) errors that are typically two to four times larger than leading hybrid functionals^{3,93}, which, in turn, have RMS errors at least ten times larger than the ‘gold-standard’ coupled-cluster theory⁹⁴ through perturbative triples (CCSD(T))⁹⁵ or even more advanced methods⁹⁶. Turning from molecules to solids, there is an even greater need for high-quality data for fine-tuning. This begins with higher-rung meta-GGA data in materials systems^{97–100}. The second error source is the choice of

We listed datasets of more than 1 million 3D structures with energy and force labels, as well as datasets of more than 100 million 3D structures without these labels. B, billion; M, million; OC20, Open Catalyst 2020; ODAC25, Open DAC 2025; OMat24, Open Materials 2024; OMC25, Open Molecular Crystals 2025; OMol25, Open Molecules 2025; SPICE, small-molecule/protein interaction chemical energies; SPF, solvated protein fragments.

At this point, it is appropriate to highlight the recent release of OMol25 (ref. 83), the largest molecular DFT dataset for training MLIPs, which covers 83 elements of the periodic table with over 100 million snapshots. In addition to generating many new structures, OMol25 encompassed configurations from the ANI-2x⁸⁸, SPICE2 (ref. 84), Transition-1x⁸⁹, OrbNet Denali⁹² and SPF⁸⁶ datasets that were recalculated at OMol25's excellent level of DFT (ω B97M-V/def2-TZVPD). OMol25 includes energies and forces for systems up to 350 atoms, up to ten unpaired electrons, total charges ranging from $-10e$ to $+10e$, large coverage of spin species, small molecules, electrolytes, biomolecules, metal complexes and intermolecular interactions, using more than ten distinct methods for structural sampling.

Error sources in DFT reference data

The main feature of pre-trained MLIP foundation models is that they can be fine-tuned with a modest amount of data to yield accurate PESs for diverse chemical systems, or extended to predict properties beyond energies and forces. At present, DFT calculations provide the best trade-off between accuracy and computational cost, such that nearly all supervised large-scale datasets for MLIPs are composed of DFT calculations. There are at least three categories of errors in the reference values of such datasets, any one of which alone could justify the need for better fine-tuning data.

The first significant source of error is the choice of functional (or wavefunctional, if one goes beyond DFT). Standard generalized gradient approximations (GGAs) have root-mean-square (RMS) errors that are typically two to four times larger than leading hybrid functionals^{3,93}, which, in turn, have RMS errors at least ten times larger than the 'gold-standard' coupled-cluster theory⁹⁴ through perturbative triples (CCSD(T))⁹⁵ or even more advanced methods⁹⁶. Turning from molecules to solids, there is an even greater need for high-quality data for fine-tuning. This begins with higher-rung meta-GGA data in materials systems^{97–100}. The second error source is the choice of

basis set. At least triple and preferably quadruple-zeta basis sets are required to converge hybrid DFT molecular relative energies towards the complete basis set (CBS) limit³. However, many of the large-scale DFT datasets summarized in Table 1 use only double-zeta basis sets (for example, 6-31G*), which induce significant errors. In condensed matter, the corresponding choice of atomic orbital basis or plane-wave cut-off is equally important^{101,102}. The third factor involves other sources of error that prevent a given ‘model-basis’ combination from yielding a well-defined value. In particular, DFT calculations on materials require careful attention to best practices^{102,103} to ensure reproducibility, beginning with choice of cut-offs, quadrature quality and convergence criteria. Such factors have been discussed and investigated for the Open Catalyst 2020 (OC20) data¹⁰⁴.

There are also other non-trivial sources of error. For example, reaching the thermodynamic limit is particularly challenging for hybrid functionals¹⁰⁵. Pseudopotential errors – typically not present in molecular calculations involving light elements – also require careful evaluation^{106–108}. Although thresholds and cut-offs also affect molecular calculations, they can be controlled and combined with self-consistent field algorithms that ensure energy descent¹⁰⁹. The local correlation methods needed for CCSD(T) on large molecules¹¹⁰ are also subject to domain errors¹¹¹. They scale with size and may reduce accuracy^{112,113} and ideally should be removed and replaced by strict numerical thresholds^{114,115}. These systematic errors make dataset fusion difficult because machine learning models trained on such datasets will probably have difficulty learning physical properties or labels from disjoint underlying distributions. In addition, peer reviewers should help find inconsistencies before initial publication, and scientists should remain committed to maintaining datasets after publication. One recent example underscoring these responsibilities is a study by Garrison et al.¹¹⁶, in which machine learning models trained on the tmQM¹¹⁷ dataset revealed that many structures in the dataset were missing hydrogen atoms. After filtering and recomputing the dataset, the model performance improved substantially.

Going beyond DFT accuracy is desirable. Here, it is appropriate to highlight the effort of Smith and co-workers in generating the ANI-1ccx dataset containing energies and forces of 500,000 organic molecules obtained with an accurate CCSD(T)* / CBS composite extrapolation^{97,118}. This protocol has deviations from the Schrödinger limit that are larger than normally acceptable for small, very-high-accuracy datasets¹¹⁹ but are much smaller than the best DFT errors, even in large basis sets³. Continued diverse collections of gold-standard CCSD(T) data (or beyond when necessary) would be extremely valuable for fine-tuning. Another result worth highlighting is the CCSD(T) assessment¹²⁰ of over 10,000 barrier heights in RDB7, which led to unexpectedly large RMS errors from the normally reliable ω B97X-D3 functional. The largest of these deviations were resolved by removing orbital instabilities and allowing the DFT to be spin-polarized when necessary¹²¹. However, this highlights the limitations of even hybrid DFT for reactive systems, particularly where strong correlation effects cause symmetry breaking¹²². Coupled-cluster benchmarks for condensed-matter periodic systems lie in the future, although there has been notable progress in coupled-cluster methods for materials science¹²³.

Priority areas for future data generation

The specific areas wherein we believe the community should devote major effort to large-scale data generation are shown in Fig. 3. For molecular DFT, in spite of recent efforts, coverage of the periodic table remains incomplete beyond organics (Fig. 3a). There is still

under-representation of transition metal–organic complexes, complex electrolytes, metalloenzymes and disordered proteins. Improving the representation of such systems could also address three other major gaps: non-covalent interactions (Fig. 3b), systems wherein long-range interactions are important (Fig. 3e), and variable charge, spin and excited states (Fig. 3c) that are poorly covered by current datasets^{124,125}. Excited-state energies and properties in particular offer an avenue for dataset development, wherein time-dependent DFT calculations¹²⁶ are relatively inexpensive (roughly the cost of ground-state DFT per state) and have been extensively benchmarked¹²⁴ against higher-level theories¹²⁷ (Fig. 3c). In addition, combustion chemistry and atmospheric chemistry, wherein different spin and excited states are also important, could be useful areas for data generation.

Reactivity, particularly involving complex mechanisms and reactions in the condensed phase, is an important area for data development (Fig. 3d). In the context of plane-wave DFT, we believe that major gaps include solid–solid interfaces (Fig. 3f), surfaces under different applied potentials (Fig. 3g), defects (Fig. 3h) and solid–liquid interfaces (Fig. 3i). We note that some of these areas will require the use of better density functionals beyond the ubiquitous PBE and RPBE. Analogous to spin states for molecular properties, additional data generation for magnetic properties could also aid in correct magnetic moment initialization for diverse materials.

Finally, we believe that there is a major need for large-scale datasets of rich quantum information, such as electron densities, orbital information, descriptors of quantum theory of atoms in molecules, and natural bond orbital interactions (Fig. 3j). OMol25 includes some electronic information, although it has only been released for a 4 million subset of the data, but no corresponding data exist for materials or any other chemical context. Infusing rich quantum information into machine learning models has recently been shown to be beneficial^{128–132}, and powerful foundation model capabilities could be unlocked with such data. For example, MLIP foundation models could be pre-trained on this extra information by swapping prediction schemes to predict atom-level and bond-level values. This could pave the way for foundational chemical models that go beyond MLIPs, serving as powerful representation models that can be adapted to predict forces and energies downstream.

Looking beyond basic properties of the MLIP, such as energies and forces, it is important to consider other physical properties used for fine-tuning foundation models. In particular, accurate spectral simulations and other response properties – which intimately depend on 3D atomic positions – would be extremely valuable, drastically expanding the properties that fine-tuned foundation models could predict. Fine-tuning atomistic foundation models with experimental data, such as radial distribution functions, solvation free energies, density and viscosity, is also desirable¹³³. We note that experimental data can also be used in the context of selecting or ‘aligning’ foundation models trained on small cluster datasets to properties better reflected by bulk simulations. For example, Gong et al. mapped MLIP forces to pressures (and, thus, densities) from a small set of representative experimental measurements¹³⁴. We discuss additional training strategies that incorporate experimental data in the following section.

Model-agnostic training strategies

Effective training strategies will continue to have an important role in developing the next generation of foundation models for chemistry and materials science. In this section, we highlight promising directions for both pre-training and post-training, with the goal of

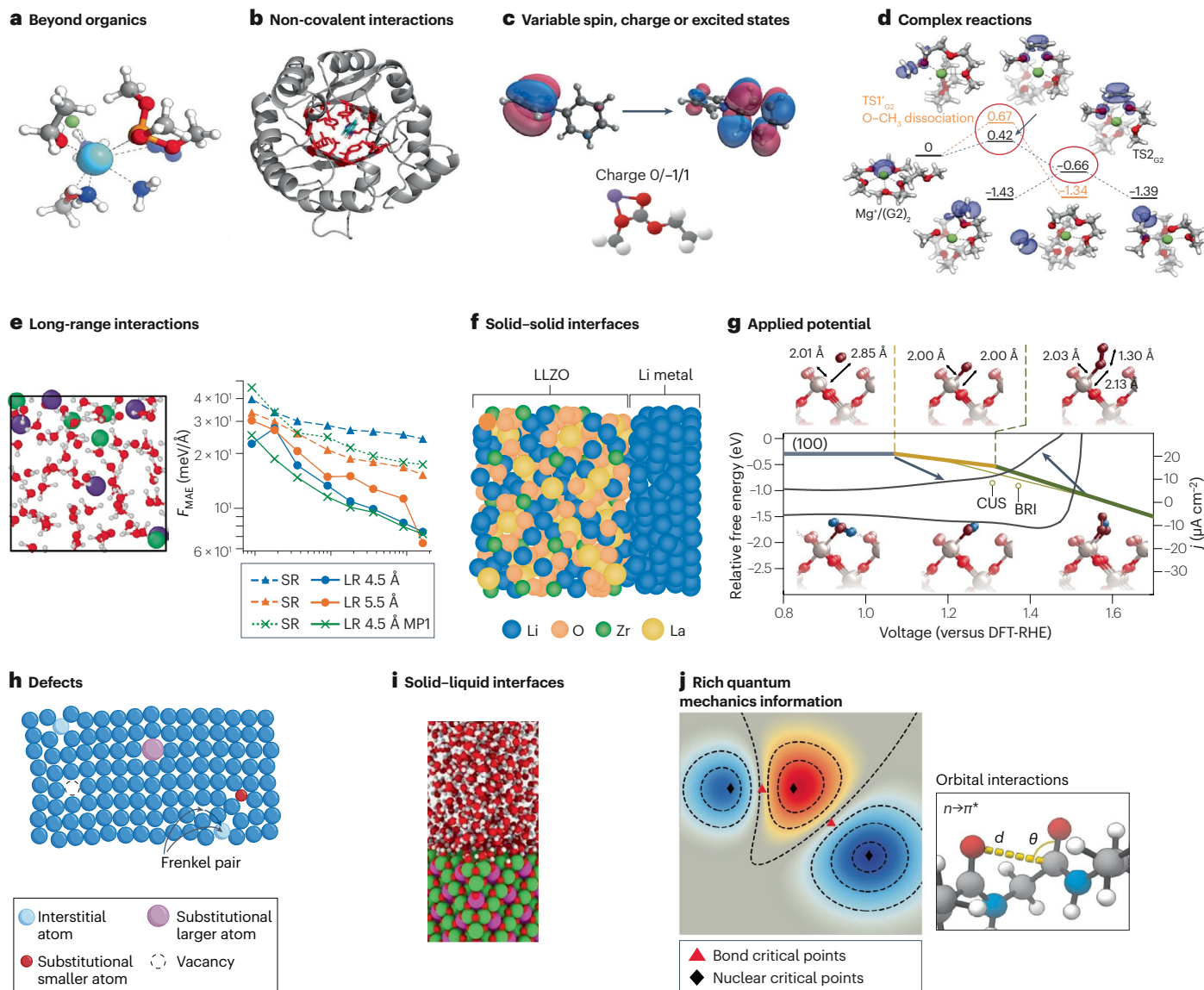


Fig. 3 | Proposed areas wherein substantial additional high-quality DFT data should be generated to realize a general and transferable foundational MLIP. **a**, Beyond-organic elements. **b**, Non-covalent interactions. **c**, variable charge, spin or excited states. **d**, Complex reactivity. **e**, Long-range interactions. **f**, Solid-solid interfaces. **g**, Surfaces under applied potential. **h**, Defects. **i**, Solid-liquid interfaces. **j**, Rich quantum information. θ , bond angle; BRI, bridging sites; CUS, coordinatively unsaturated sites; d , interatomic distance; DFT, density functional theory; F_{MAE} , force mean absolute error; j , current density; LLZO, lithium lanthanum zirconium oxide; LR, long range; MLIP, machine-learned interatomic

potential; RHE, reversible hydrogen electrode; SR, short range; TS, transition state. Parts **a–c** are adapted with permission from refs. [124](#),[200](#),[201](#), respectively, ACS. Part **d** is reprinted with permission from ref. [202](#) under a Creative Commons licence [CC BY 4.0](#). Part **e** is reprinted from ref. [71](#), Springer Nature Limited. Part **f** is adapted with permission from ref. [203](#), Springer Nature Limited. Part **g** is reprinted from ref. [204](#), Springer Nature Limited. Part **i** is reprinted from ref. [34](#) under a Creative Commons licence [CC BY 4.0](#). Part **j** (left) is reprinted with permission from ref. [129](#), RSC, whereas part **j** (right) is reprinted from ref. [131](#), Springer Nature Limited.

obtaining ‘model-agnostic’ training strategies that can be applied to any foundational MLIP.

Pre-training strategies

Unsupervised training methods have been crucial in improving the foundational capacity of LLMs and LVMs. However, in computational chemistry and materials science, atomistic configurations are typically

associated with energy and/or force labels (or molecular property labels, for example, in Zinc22 (ref. [78](#))/Zinc20 (ref. [77](#))), making supervised learning the common approach for MLIP optimization. By contrast, self-supervised approaches start with an unlabelled data distribution, which is often centred on molecular geometry. Examples of this are the PCQM4Mv2 (ref. [135](#)) and PM6 (ref. [136](#)) datasets developed under the PubChemQC Project¹³⁷. In one recent study,

Ji and co-workers demonstrated that training with a self-supervised coordinate unmasking and denoising objective on 800 million unlabelled 3D conformations of organic molecules improved downstream performance on QM9 property prediction⁷⁶. Another recent work has used geometric denoising to create a pre-trained model that exhibited fine-tuned accuracy improvement compared with randomly initialized ‘from scratch’ models, also demonstrated for various chemical properties in the QM9 dataset^{56,138}. This denoising approach was later extended to work for non-equilibrium structures¹³⁹. In a similar vein, the recent ORB model was pre-trained as a denoising diffusion model¹⁴⁰ before being fine-tuned as a MLIP⁵⁶. Understanding the extent to which the success of self-supervised pre-training relies on the quality of the training distribution, and how scalable such approaches are relative to purely supervised (pre-)training, is an active area of investigation. Another approach gaining traction is pre-training using synthetic energy and force data provided by other MLIPs trained from scratch. This has been demonstrated to improve the robustness of fine-tuned MLIPs given that the synthetic source is reasonably reliable¹⁴¹. Other examples of self-supervised learning approaches in materials chemistry include twin algorithms¹⁴² and atom replacement¹⁴³ pre-training schemes.

Although most current models train on a dataset with calculations performed at a single level of theory^{34,144}, future foundational MLIPs could benefit from training on multiple levels of theory. A training objective that extracts learning signals from multiple levels of theory could take advantage of the abundant datasets generated using cheaper methods (for example, semi-empirical potentials), while still maintaining accuracy by leveraging higher-fidelity data. Yuan and Head-Gordon have recently reported a pre-training scheme that uses effectively free, single-molecule non-reactive force field data, while all intermolecular interactions and reactive properties are learned at a fine-tuning stage with a small amount of computationally more expensive labels compared with models trained from scratch¹⁴⁵ (see Fig. 4a–c). Incorporating different levels of theory into the training pipeline would also allow MLIPs to learn different interactions best described by different methods^{32,146,147}. However, training a model that can handle different energy scales and the nuances of each level of theory remains difficult. Addressing these challenges could lead to the development of more general MLIPs.

Physical laws can also be incorporated into MLIPs by placing them directly into the loss function and training procedure, instead of making them inherent to the model architecture. For example, rotational equivariance can be enforced via data-augmentation loss, as discussed previously, and energy conservation can be encouraged by minimizing the curl of the MLIP as part of the training objective¹⁴⁸. Although these ‘soft’ constraints do not guarantee that the law will be obeyed, they circumvent the need to design new architectures for each new constraint of interest, and they enable more scalable and flexible training¹⁴⁹.

Post-training strategies

After pre-training a foundation model, post-training is important for further refining and optimizing the model to a given downstream application. For domain-specific tasks, such as energy and force prediction, modest amounts of higher-quality and labelled data are used within a standard supervised fine-tuning context. Model distillation¹⁵⁰ is another post-training approach in which a simpler model (the ‘student’) is trained to replicate the behaviour and performance of a larger, more complex model (the ‘teacher’, or foundation model in this case). This process involves transferring knowledge from the teacher

to the student by using the predictions or internal representations of the teacher as targets for training the student model. The use of teacher–student constructs frequently allows more computationally efficient training to be performed, particularly for multi-label training.

The effectiveness of model distillation has recently been demonstrated in several studies. For example, Kelvinius et al. performed MLIP distillation by aligning node and edge features across models, but they did not consider the task of specializing from general-purpose foundation models¹⁵¹. In another example, Amin and co-workers¹⁵² used distillation on large-scale models such as MACE-OFF¹⁴⁴, MACE-MP-0 (ref. 34) and JMP¹⁴⁶. By matching the Hessians of the energy predictions, these large-scale models were distilled into small student MLIPs specialized for a specific chemical subset of the data. These models were up to 50 times faster during inference time than the original large-scale model (Fig. 4d). Notably, this distillation approach is agnostic to the MLIP architecture and is compatible with arbitrary combinations of teacher and student MLIP design choices. Distillation also offers a potential solution to balance scalable training while ensuring physical consistency in the final simulation. For example¹⁵², a large-scale model can be scalably trained without expensive constraints (such as gradient-based forces for energy conservation) and can then be distilled into a student MLIP possessing these constraints. This ensures energy conservation while leveraging the general-purpose representations from the pre-trained large-scale model. Both pre-training and post-training strategies could enable many enhanced downstream tasks as shown in Fig. 4e, ranging from geometry optimization to modelling reactive trajectories.

Thus far, much of the discussion has been about MLIPs trained on *ab initio* data. However, experimental data is also a rich potential source of information to train MLIPs. Directly incorporating experimental data into MLIP training has primarily been done through differentiable simulation^{133,153–157}. In this case, MLIPs are typically initialized by pre-training with an analytical prior potential, or on a sparse set of simulated observables (such as radial distribution functions, phonon densities and diffusivity coefficients), which can be computed as ensemble averages over an equilibrium molecular dynamics simulation. The pre-trained model is then fine-tuned to match the experimental observables. Computing gradients necessary for MLIP fine-tuning through the full molecular dynamics simulation can be prohibitively expensive. Therefore, techniques such as reweighting¹⁵⁴, implicit differentiation^{133,158} and the adjoint method¹⁵⁹ are used to enable tractable optimization. However, major technical challenges remain, the most notable of which is effectively training with ‘dynamical’ observables¹⁶⁰.

Evaluation of a successful foundation model

Although competition in machine learning model performance on small datasets such as MD17 (ref. 148) drove crucial early MLIP development, large community efforts have recently emerged that define a successive series of ‘leaderboards’ as testing grounds for MLIPs. Some of these are focused narrowly on energy and force errors, whereas more recently, assessment has broadened to downstream application success, such as stable molecular dynamics trajectories, exhibiting scaling laws, and even the possibility of emergent behaviour.

Criteria for foundational MLIPs

Scaling laws help us understand the emergent generalization capabilities of LLMs. Hence, we claim that MLIP foundation models should comply with heuristic scaling laws, with the expectation that they too

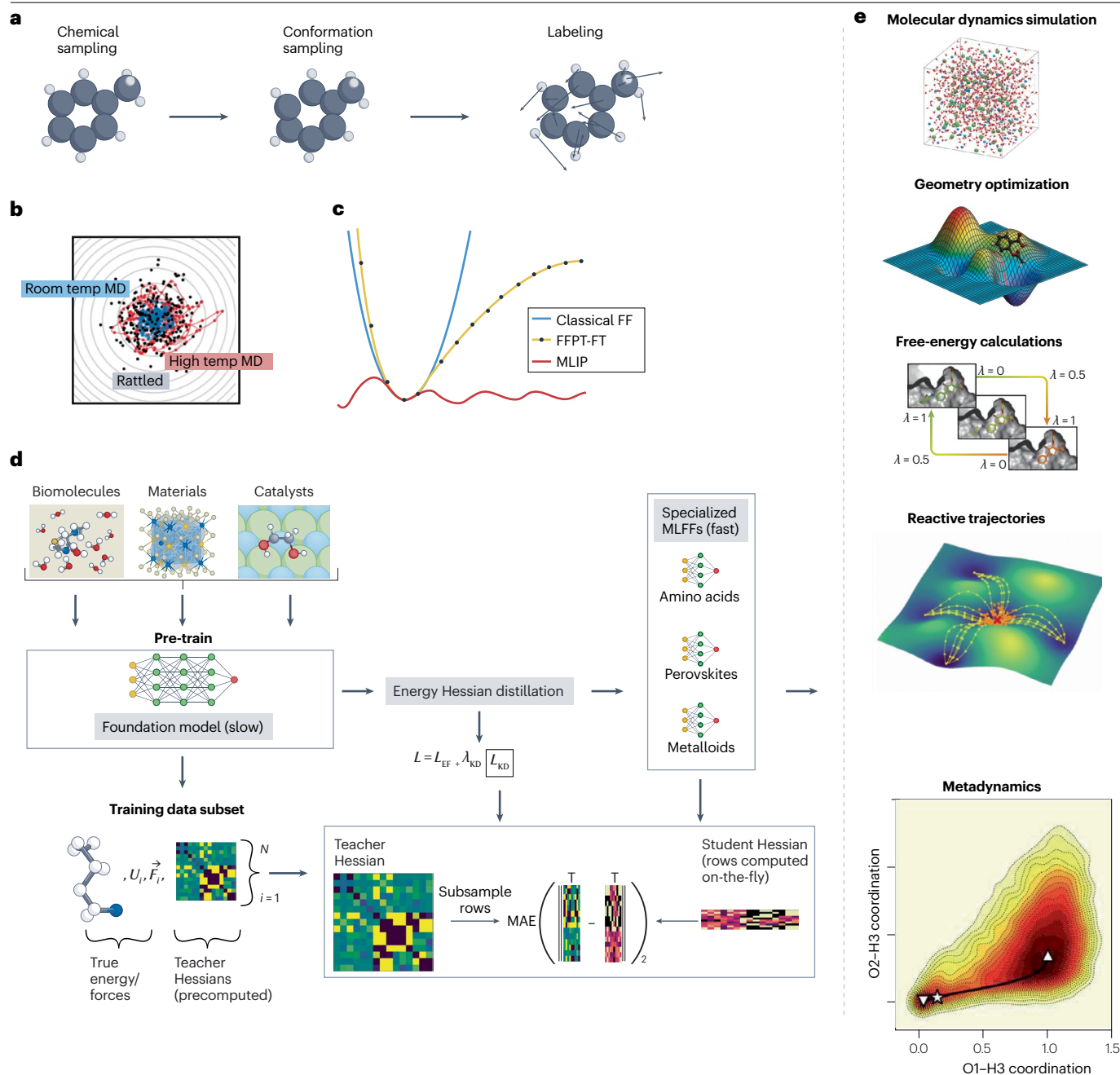


Fig. 4 | Examples of pre-training and post-training strategies for machine learning interatomic potentials. **a**, Pre-training strategy of sampling high energy and unphysical data using rattling and cheap force field labelling for pre-training a MLIP with subsequent fine-tuning. **b**, Compared with accumulating frames from ab initio molecular dynamics (AIMD) simulations at high temperatures (grey curve), rattling (blue curve) generates more uncorrelated structures. **c**, The PES from an MLIP trained on chemically relevant data (red curve) tends to be rough and incomplete and on a lower energy scale, leading to simulation instabilities. The PES from FFPT (yellow curve) has the correct limiting behaviour despite its lower accuracy, but it can be fine-tuned for a more complete PES with high-quality data (blue curve). **d**, Post-training model distillation starts with an MLIP foundation model that has been trained on a large quantity of

diverse data. A series of smaller MLIPs can then be trained via a knowledge distillation procedure, enabling specialization and speed for a specific task while retaining the general-purpose representations learned by the large-scale foundation model. **e**, Examples of downstream tasks enabled by pre-training and post-training strategies. λ_{KD} , hyperparameter controlling relative contribution of L_{EF} and L_{KD} ; DFT, density functional theory; F , forces; L , loss function; L_{EF} , ground-truth loss; L_{KD} , knowledge distillation loss; MAE, mean absolute error; MD, molecular dynamics; MLFF, machine-learned force field; MLIP, machine-learned interatomic potential; N , Hessian dimension; PES, potential energy surface; U , potential energy. Parts **d** and **e** are reprinted from ref. 152 under a Creative Commons licence [CC BY 4.0](https://creativecommons.org/licenses/by/4.0/).

would perform well on a diverse range of tasks beyond just energies and forces after fine-tuning and post-training. Downstream tasks might include predicting different levels of *ab initio* theory, types of chemical or periodic systems, or field-dependent properties, serving as generative models for crafting new structures or molecules^{63,138,141,142,161}, measuring outcomes for defective structures^{162,163}, modelling excited state potential energy surfaces, and predicting dynamical quantities through molecular dynamics simulations under more varied conditions^{17,163–165}.

There are a few prototype MLIP foundation models that show the ability to comply with heuristic scaling laws. In particular, pre-trained models, such as Efficiently Scaled Attention Interatomic Potential (EScAIP)⁴⁵, Uni-Mol2 (ref. 76) and the Universal Model of Atoms (UMA) pre-trained MLIP¹⁶⁶, have characterized the scaling correlations of decreasing validation error with increasing model size, dataset size and computational resources. This suggests that emergent capabilities might follow, such as predicting energy labels from unlabelled data, as shown for Uni-Mol2 (ref. 76). An exciting new development is the release of UMA, based on the eSEN architecture¹⁶⁷ and is trained on nearly 500 million samples – the largest model and most diverse training set ever used to train a single MLIP. Although UMA may come closest to our vision of a foundation model for atomistic simulation, it is crucial to develop testing platforms for MLIP foundation models to fully test their endowed capabilities.

Evaluation platforms

The OC20 leaderboard¹⁶⁸ was established in 2021. Its dataset is split into training, validation and testing categories with rigorously defined in-distribution and out-of-distribution (OOD) systems, and a private test set to prevent test performance from being gamed. Since its inception, the performance of energy and force models has consistently improved. In 2023, the MatBench Discovery leaderboard⁵⁹ (Fig. 5a) was created to test MLIP performance for the relaxations of crystalline materials and the ability to predict the thermodynamic stability and formation energy of materials. Although MatBench Discovery was initially only used for models trained on MPtrj³⁶, interest and adoption has since widened. Recently, teams from Google, Microsoft and Meta each reported the generation of supplemental training data, as well as models trained on their data beating the previous best performing model.

Leaderboard evaluations are not without limitations. For example, introducing new training data may risk leakage (that is, overlap) with the test set of a leaderboard. Furthermore, most of the leaderboard evaluations are overly focused on energy predictions, which are less diagnostic for widely used workflows such as molecular dynamics simulations. A related issue is the lack of robust uncertainty quantification, which has not been taken into account in existing leaderboards. The prevalent method for quantifying uncertainty in MLIP predictions involves using ensembles^{169–171}; this approach entails running multiple fits of the MLIP simultaneously and interpreting the variances among them as a measure of uncertainty. This is computationally demanding, requires parallelization and may be unsuitable for large-scale MLIP foundation models. Both issues have consequences for deploying MLIPs in practical applications, and more rigorous tests of uncertainty quantification should be incorporated into future MLIP benchmarks.

Even if the MLIP universality within a specific domain can be evaluated through OOD tests, current leaderboards do not measure whether the resulting models are truly foundational – that is, that they obey scaling laws, are derived from diverse data or pre-training strategies, and exhibit superior performance across a broad range of downstream

tasks after fine-tuning. For example, although the large size of the OC20 training set makes it viable for evaluating model scalability (for example, via performance with 1M vs 10M vs 100M training data points), only performance on the full training dataset is featured on the leaderboard. Furthermore, both OC20 and MatBench Discovery focus on well-behaved relaxations but without consideration of other emergent properties, such as molecular dynamics stability or PES smoothness, which has been shown not to be necessarily correlated with test errors on energy and force prediction^{161,164,172,173}.

The recently released NNP Arena¹⁷⁴ is a promising development that addresses some of these limitations (Fig. 5b). It evaluates energy error for both pre-trained MLIPs (and pure DFT models) against gold-standard coupled-cluster reference values, decoupling it from the MLIP training procedure and data. It also evaluates inference speed, which previous leaderboards have not considered. However, the GPU hardware on which these comparisons are performed is currently not standardized, which is an important consideration for the future. The recent MLIP Arena¹⁷⁵ (Fig. 5c) expands the scope of MLIP benchmarking by assessing the extent to which models comply with fundamental physics, including energy conservation, asymptotic behaviour, smoothness, symmetry and simulation stability. These metrics do not require additional labels, yet they carry substantial implications for the reliability and applicability of the models in practical settings.

To illustrate progress in MLIP development as measured by these evaluation platforms, UMA and eSEN-OMol now top Rowan's NNP Arena leaderboard for neutral, closed-shell, organic molecular energies, with errors against CCSD(T) for the GMTKN55 benchmark set falling by nearly 5 kcal mol⁻¹ compared with the previous best model¹⁷⁴. Digging more deeply into the GMTKN55 test set¹⁷⁶, one can see that the energies of UMA are already of lower error with respect to CCSD(T) than those of ω B97M-D3BJ/def2-QZVP DFT for variable charge, variable spin and metal-containing large systems and barrier heights, while being nearly comparable for intermolecular non-covalent interactions. However, the OMol25 evaluation tasks (Fig. 5d) demonstrate that there is still substantial room for improvement on charge, spin and long-range interactions, wherein error against reference DFT data remains very large (>100 meV) probably owing to the naive handling of charge and spin and the 6 Å graph cut-off, emphasizing the key areas wherein further MLIP architecture development is needed. In addition to the >100 million training data, OMol25 also includes explicit out-of-distribution test sets and novel model evaluations, which now populates a public leaderboard¹⁷⁷.

Future directions

MLIP foundation models promise to enable massively impactful investigations of chemical and materials systems that are beyond the reach of current physics-only models, while avoiding the need to build each machine learning model from the ground up for any new physical system of interest. To illustrate this, consider that whole-system electrochemical and catalysis modelling under *operando* conditions is currently bottlenecked by short computational timescales, too many environmental degrees of freedom, long-timescale dynamics, and non-equilibrium phenomena. The advent of foundational MLIPs could overcome these time and length scale issues for ground-state potential energy surfaces. Additionally, efficient and cost-effective methods for storing and converting energy, such as artificial light harvesting, fuel cells and batteries, are inherently excited-state quantum phenomena that are beyond the capabilities of our best physics models. Therefore, foundational MLIPs for excited states will expand the space of new materials relevant

Perspective

to photocatalysts and solar-energy storage. Finally, although almost 100,000 molecular organic frameworks (metal–organic frameworks (MOFs), and related covalent versions (COFs)) have been synthesized to date, there is little understanding behind their assembly, and hence

they rely on brute-force high-throughput screening and trial and error. The combined use of LLMs with foundational MLIP workflows could dramatically increase the pipeline for discovery of new MOFs and COFs for hydrogen storage or water capture.

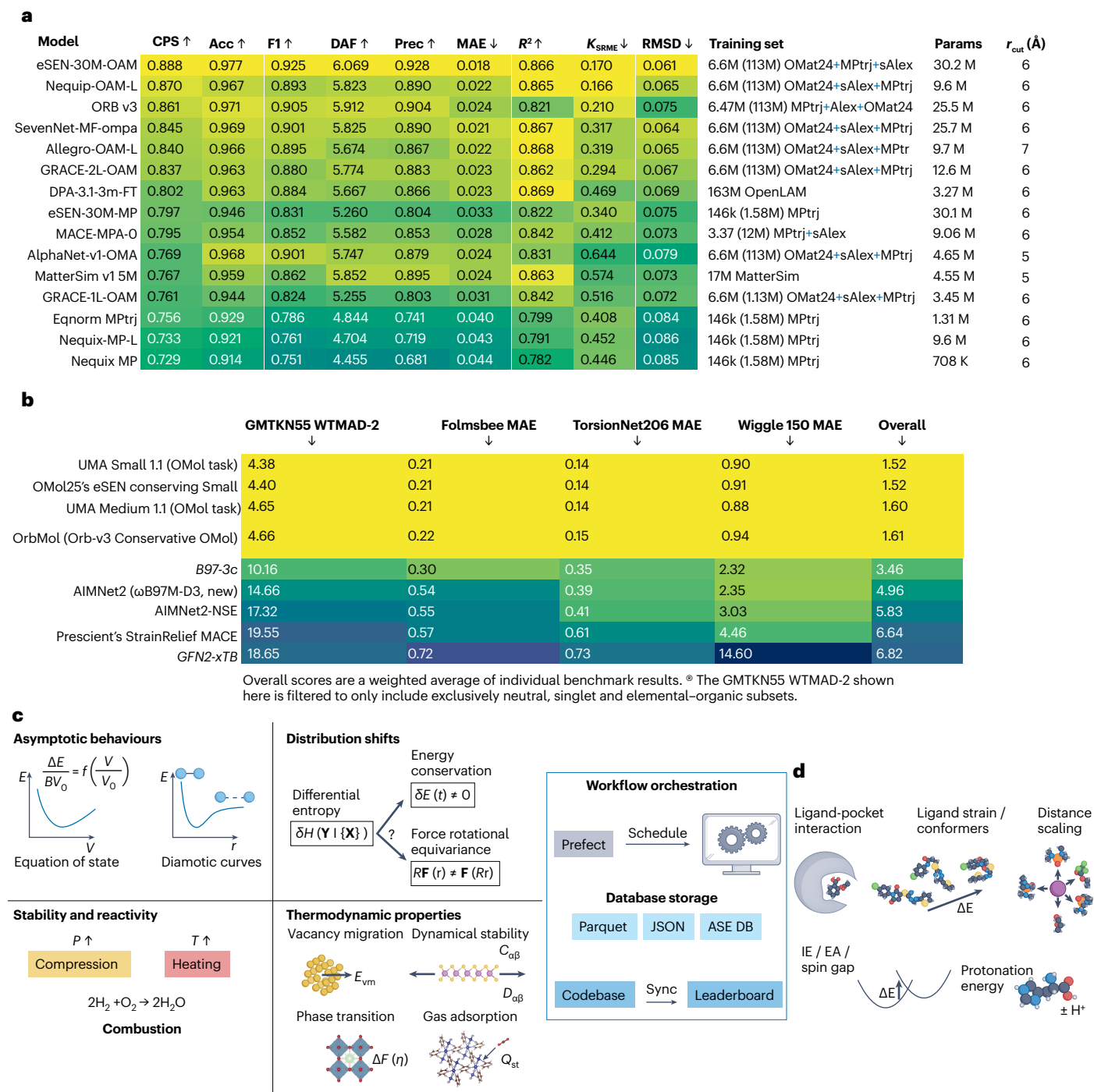


Fig. 5 | Evaluation platforms used to assess MLIP performance. MatBench Discovery (part a) and Rowan (part b) leaderboards keep a running assessment of numerical rankings of MLIP performance based on community submissions. MLIP Arena (part c) and OMol25 (part d) provide assessment of important physical behaviour of MLIPs that cover a range of materials and chemical

properties. EA, electron affinity; IE, ionization energy; K_{SRME} , SRME in lattice thermal conductivity; MAE, mean absolute error; MLIP, machine-learned interatomic potential; RMSD, root-mean-square deviation; SRME, symmetric relative mean error. MLIP Arena is adapted from ref. 175 under a Creative Commons licence CC BY 4.0.

A key insight gained from other areas of machine learning and artificial intelligence is that methods that may work in ‘low-data’ regimes do not necessarily work as well in higher-data regimes. Currently, the two paradigms coexist: one uses inductive biases to address physical constraints (which often requires less data)^{42,43}, whereas the other states that physical laws can be learned implicitly when there is enough data, and when model capacity and compute is correspondingly scaled to take advantage of this data. In this Perspective, we have highlighted ways in which these paradigms can be effectively merged. Training MLIPs through constraints and learning energy and forces together ensures that conservative forces are available for the primary downstream molecular dynamics simulations¹⁴⁹. However, post-training strategies can take non-conservative MLIPs and distill them into conservative MLIPs¹⁵², enabling both initial training at scale and physical soundness at deployment. Hence, we posit here that incorporation of architectural inductive biases is not the only way to create sound MLIP foundation models, a position that is not yet fully reflected in the recent literature. More broadly, whether the chemical and materials sciences will escape the bitter lesson, or whether we will eventually also conclude that data and scalability is superior to explicitly built-in constraints, as has come to pass in other fields grounded in physics, such as robotics and weather forecasting, remains an interesting open question.

Training a truly foundational MLIP will require very large datasets that broadly span chemistry, materials, phases and interfaces. We focused our earlier discussion on the use of DFT-labelled data as the standard approach for pre-training. New datasets are being released at an astounding rate, including OMol25 (ref. 83) for molecules, the Open DAC 2025 (ODAC25) dataset¹⁷⁸, a significant expansion and improvement upon ODAC23 (ref. 179) for molecular gas (CO₂, H₂O, N₂ and O₂) adsorption in 15,000 MOFs, and OC25 as the successor to OC20 and OC22, spanning 88 elements, commonly used solvents or ions, varying solvent layers, and off-equilibrium sampling¹⁸⁰. However, much larger low-level datasets (with or without labels) from classical force fields, semi-empirical methods, or on-the-fly schemes for computing properties could be important to unlock powerful large-scale capabilities for chemical foundation models. It is also possible that the best approach will not actually start with an MLIP but will instead leverage some combination of traditional chemoinformatics, chemically relevant text, and/or unlabelled structures to create datasets suited for larger foundation models that can be retooled as MLIPs. Information contained in other multimodal data that incorporates experimental information could also be extremely valuable for pre-training. However, in practice, this will require extensive data curation to reach a data scale suitable for pre-training, as well as the development of novel training strategies (even beyond what we have discussed) that are viable at scale. The fine-tuning of foundational MLIPs for the highest fidelity in specific applications will require data accuracy that goes beyond the limits of present-day DFT^{3,181}, wherein a single semi-local or hybrid functional cannot yet span molecules to materials with equal accuracy¹¹⁹.

To achieve rapid turnaround (that is, high speed and efficiency of training, fine-tuning and evaluation), efficient mechanisms for storing and retrieving data for batched training, dynamic load balancing methods that can accommodate variations in graph sizes, and communication-hiding techniques that overlap computation and message passing will continue to be important. In addition, the development of a scalable foundation model will require the use of efficient algorithms to perform pre-training and fine-tuning. For example, UMA

is trained on OMol25 (ref. 83), OMat24 (ref. 62), ODAC23 (ref. 179), OC20 (ref. 104) and the novel Open Molecular Crystal dataset¹⁸², and it uses a mixture of linear experts (MoLE) for two key benefits¹⁶⁷. First, MoLE allows training in a multi-task fashion, that is, each distinct training dataset is a ‘task’, and all tasks are trained together, which results in superior performance relative to task-specific training. Second, MoLE allows a huge number of learnable parameters to be active during training, but a much smaller number to be active at inference time, allowing the model to effectively leverage the vast training dataset while still remaining fairly fast and memory efficient for applications. We anticipate that further advancements in this vein will be necessary to make MLIPs practical in downstream applications.

It remains important to develop testing platforms for MLIP foundation models to quantify their endowed capabilities. This will have considerable computational cost, owing to the large number of simulation runs required, as well as the need for studies to validate testing components and analyses to derive scientific insights. Nevertheless, we believe that the benefits to the scientific community are profound and represent an essential direction for future research in demonstrating the promise of MLIP foundation models^{164,183–185}. Although current MLIP testing platforms are a valuable resource for the community, there are significant challenges in how MLIP foundation models are evaluated and the mechanisms needed for training and fine-tuning at scale. For example, there is not always a direct correlation between reduced force errors observed in test set performance and the actual performance in molecular dynamics simulations^{161,164,172,173}. Current MLIP leaderboards also do not yet explicitly evaluate model scalability with respect to the amount of training data or learnable parameters, nor do they seek to quantify the ‘foundational capacity’ of the model via standardized fine-tuning across chemical and materials domains and properties. With scaling laws being the defining property of a foundation model, MLIPs should demonstrate continuous improvement with more parameters, more data and more compute. Ultimately, the practitioner must demonstrate the success of a foundation model via fine-tuned models that are fast and specialized and that are more accurate than a model trained from scratch on task-specific data.

The success of foundational LLMs and LVMs has relied on scraping vast collections of text and images from the internet – data obtained via implied or enforced open access. However, in the context of *ab initio* data for training MLIPs, this assumption of open access to data is not always adhered to when commercial players are involved. For example, the datasets used to train Google’s GNoME model for crystal energies and Microsoft Research’s MatterSim have not been publicly released. By contrast, the FAIR Chemistry team at Meta, which is responsible for the largest MLIP training datasets built to date, has consistently released their datasets (and trained models) fully open source – a contribution to the scientific community that cannot be understated. In a related vein, MLIP models have benefited tremendously from industry-released software tools, such as TensorFlow¹⁸⁶, Keras¹⁸⁷ and PyTorch¹⁸⁸. However, open-access development has not occurred for LLMs such as ChatGPT¹⁸⁹, and the loss of implementation details on how LLM foundation models work could limit our ability to develop similarly powerful MLIP foundation models.

Ethical, regulatory, legal and practical issues must also be considered for the development of MLIP foundation models. We note that although DeepSeek-V3 is open source and can deliver high-quality results at a lower cost compared with close-source LLMs, it is either restricted or outright banned in several countries because of concerns

Glossary

Active learning

An iterative strategy for selecting informative configurations for labelling to improve an MLIP after many rounds of retraining.

Atomic cluster expansion

(ACE). A framework for constructing atom-centred descriptors using systematically improvable body-ordered terms, which helps in describing atomic environments in machine learning interatomic potentials.

Attention

A mechanism used in state-of-the-art machine learning models that dynamically weighs different input elements based on their relevance to a given task, enabling models to focus on the most important information.

Born–Oppenheimer approximation

An approximation in quantum chemistry that separates the motion of nuclei and electrons in a molecule, simplifying the Schrödinger equation.

Compute scaling

The ability of a system to handle more users, data or load without losing performance.

Coupled cluster

A method in computational chemistry to describe the electronic structure of many-electron systems, often regarded as the gold-standard of accuracy in quantum chemistry.

Density functional theory

(DFT). A method in computational chemistry to describe the electronic structure of many-electron systems, often regarded as an affordable and sufficiently accurate choice in quantum chemistry.

Direct force models

Models that directly predict atomic force vectors without computing the gradient of a scalar potential energy, thereby accelerating model inference while sacrificing their connection with the energy and the zero-curl property necessary for energy-conserving simulations.

Equivariance

A property of a model wherein its predictions and/or representations undergo the same changes under transformations, in particular rotations, of the input data, important for ensuring physical consistency in simulations.

Fine-tuning

A process wherein a pre-trained model is further trained, often on a labelled dataset with task-specific supervision, to improve its performance on a target application.

Foundation model

A large-scale model pre-trained on diverse data to capture complex patterns, which can be fine-tuned for various downstream tasks.

Gradient-based force model

A model that predicts atomic force vectors as the gradient of a scalar potential energy, thereby guaranteeing the conservation of energy in chemical simulations.

Graph neural networks

A type of neural network designed to process data structured as graphs, wherein nodes represent entities and edges represent relationships between them.

Inference

The use of a machine learning model to make predictions on new data not seen during training.

Invariances

Properties of a model wherein their predictions and/or representations remain unchanged under transformations, in particular translations and rotations, of the input data, important for ensuring physical consistency in simulations.

Machine-learned interatomic potentials

(MLIPs). Models that use machine learning techniques to predict the potential energy surface of atomic systems, enabling simulations of molecular dynamics, geometry optimizations and other downstream applications.

Message passing

A process in graph neural networks wherein information is exchanged between nodes to capture relationships and interactions in the data.

Model distillation

A technique wherein a simpler model is trained to replicate the behaviour of a larger, more complex model, transferring knowledge from the teacher to the student model.

Potential energy surface

(PES). A property in computational chemistry that defines the electronic energy as a function of nuclear positions under the Born–Oppenheimer approximation.

Pre-training

A stage in machine learning wherein a model is trained on a large dataset to learn general-purpose representations.

Scaling laws

Empirical rules that describe how the performance of a model improves with increasing model size, training data and computational resources.

Self-supervised learning

A type of machine learning wherein models are trained on unlabelled data by predicting parts of the input from other parts, often used for pre-training large models.

Supervised learning

A type of machine learning wherein models are trained on labelled data by predicting a pre-calculated output from the input.

Training

The process by which the parameters of a machine learning model are learned, typically via gradient-based optimization of an objective function averaged over a large quantity of data.

Transfer learning

A machine learning technique wherein a model developed for one task is reused as the starting point to train a model for another distinct but relevant task.

Uncertainty quantification

A technique used in machine learning to estimate confidence in model predictions, helping to assess reliability and robustness in decision-making.

Universal potential

A machine learning model trained to predict potential energies and forces across a wide range of chemical systems instead of a single specific system.

about ethics, privacy and security. As for all emerging technologies, training of researchers and practitioners in ethical development and use of artificial intelligence or machine learning should remain a top priority in the physical sciences¹⁹⁰. A notable practical challenge is the massive computational resources required to demonstrate

foundation model capabilities at a large scale. This demands world-class, high-performance computing beyond typical academic laboratory facilities. An obvious first place to turn for resourcing at this scale is the leadership-class computing facilities at national laboratories. The combined issues of open-source data, software, models and ethics training

will probably require a large federation of government agencies and industry partners, with strong international cooperation – modelled on the large-scale cooperation that led to the success of the Human Genome Project¹⁹¹.

Published online: 11 February 2026

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Acknowledgements

The authors thank E. Qu, Y. Wang, J. Cavanagh, K. O. Sun, K. Hegazy, J. P. Heindel, P. Zhong, A. Kumar, S. Liang, S. Sami, R. A. LaCour, S. K. Chandy, R. Zhao and D. Bagni for their contributions in discussions on 6 August 2024 at UC Berkeley. T.H.-G. thanks the CPIMS programme, Chemical Sciences Division, Office of Basic Energy Sciences, Office of Science of the US Department of Energy under contract DE-AC02-05CH11231 for support of the machine learning research. S.M.B. acknowledges support by the Center for High Precision Patterning Science (CHiPPS), an Energy Frontier Research Center funded by the Basic Energy Sciences, Office of Science of the US Department of Energy at Lawrence Berkeley National Laboratory under contract DE-AC02-05CH11231. S.V. was supported by the Office of Defense Nuclear Nonproliferation Research and Development (NA-22), National Nuclear Security Administration, US Department of Energy, as part of the NextGen Nonproliferation Leadership Development Program. This work was also supported by the Office of Advanced Scientific Computing and Office of Basic Energy Sciences, Office of Science, US Department of Energy, via the Scientific

Discovery through Advanced Computing (SciDAC) programme for the mathematical and computing foundations of machine learning models. This work was also supported by the Energy Earthshot initiatives of the Office of Science, US Department of Energy, as part of the Center for Ionomer-based Water Electrolysis at Lawrence Berkeley National Laboratory under award number DE-AC02-05CH11231, as well as Toyota Research Institute as part of the Synthesis Advanced Research Challenge. This work used computational resources provided by the National Energy Research Scientific Computing Center (NERSC), a US Department of Energy Office of Science user facility operated under contract DE-AC02-05CH11231.

Author contributions

S.M.B., B.C., A.K. and T.H.-G. defined the topics and scope of the Perspective. E.C.-Y.Y., Y.L., J.C., S.R., T.K., S.V. and W.X. made the figures. E.C.-Y.Y., Y.L., J.C., S.R., T.K., S.V., P.Z., W.X., M.H.-G., S.M.B., B.C., A.K. and T.H.-G. wrote the paper. All authors discussed the results and made comments and edits to the manuscript.

Competing interests

The authors declare no competing interests.

Additional information

Peer review information *Nature Reviews Chemistry* thanks Sergei Tretiak and the other, anonymous, reviewer(s) for their contribution to the peer review of this work.

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